# Leonardo Tonelli

leonardotonelli03@gmail.com | LinkedIn | Personal Website | GitHub

## EDUCATION

**EPFL - École Polytechnique Fédérale de Lausanne** *Lausanne, Switzerland*
**Title:** MSc. in Statistics *Sept 2025 - Present*
**GPA:** 5.5/6.0
**Coursework:** Statistical Computation, Statistical Inference, Math of Data, Statistical Machine Learning, Regression Methods

**BOCCONI UNIVERSITY** *Milan, Italy*
**Title:** BCs in Economics and Computer Science *Sept 2022 – July 2025*
**Final Mark**: 110/110 Cum Laude
**Thesis:** *"Replicated Approach to the Binary Perceptron"* (supervised by Prof. Luca Saglietti)
**Coursework**: Advanced Statistics, Econometrics, Applied Mathematics, Machine Learning, Computer Programming

**NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE** *Singapore,*
**Title:** Exchange Student in Business & Computing programme *Jan 2025 – May 2025*
90th percentile over 1179 exchange undergraduate winning applicants' scores.
**Coursework**: Business Analytics consulting (A+), Signal and Noise in Biology (A+) and Intelligent Agents (A+)

## PROFESSIONAL EXPERIENCES

**RESEARCH ASSISTANT –** *Bocconi University* *Milan, Italy*
*Sept 2024 – Dec 2024*

- Analyzed RCT data (~1,200 participants; ~1M eye-tracking observations) to estimate treatment effects using ANOVA and regularized GLMs
- Built reproducible Python pipelines for large-scale behavioral data preprocessing and model validation
- Collaborated cross-functionally to translate statistical findings into policy-relevant insights

**AI SOLUTIONS DEVELOPER** – *Obloo Ventures, Venture Capital and Private Equity* *Milan, Italy*
*Jun - Sept 2024*

- Developed NLP application for audio transcription, summarization, and topic extraction.
- Implemented and tested RAG architecture; boosted efficiency by 70%, increased value extraction by 3x
- **Technologies**: Docker, Streamlit, Groq, Hugging Face, Git

## SELECTED PROJECTS

### Bachelor Thesis – Replicated Algorithms for the Binary Perceptron

Reproduced entropy-based learning algorithms from Baldassi et al. for the Binary Perceptron (NP-hard constraint satisfaction problem) using statistical mechanics methods; derived and implemented Replicated Simulated Annealing and Replicated Gradient Descent to explore dense solution regions of the perceptron loss landscape, and conducted experiments across problem sizes ($N \leq 500$) and constraint densities ($\alpha \leq 0.5$), demonstrating improved convergence over non-replicated baselines and analyzing phase-transition behavior in solvability.

### Deep Q-Network Reproduction – Atari Reinforcement Learning

Led a team of 4 to reproduce the Deep Q-Network (Mnih et al., 2013) in PyTorch, implementing experience replay, target networks, reward clipping, and ε-greedy exploration; trained a CNN-based agent on Atari Demon Attack for 1,000 episodes and analyzed learning dynamics via Q-value and reward convergence, validating policy behavior through post-training simulations and documenting implementation and theoretical foundations.

## TECHNICAL SKILLS

**Python**: NumPy, SciPy, Pandas, PyTorch, scikit-learn, statsmodels, Seaborn, Matplotlib, Spacy | **R**: Statistics (30/30), Econometrics (28/30) | **SQL**: 30 cum laude/30 Bocconi exam | **LaTeX**: academic experience | **Git**: professional experience

## HONORS & AWARDS

- **2024:** MEF Welfare Fund Scholarship (4000€). Merit-based award for 1st and 2nd year university academic results
- **2022:** Register of Excellence, Italian Ministry of Education, Top 3.4% nationally in final high school exams
- **2020:** Olimpiadi di Italiano (Italian Olympiad) – National semifinalist
- **2018:** INPS Programma Itaca Scholarship (€6,000) – Ranked 27th out of 737 applicants nationwide for study abroad program; spent one semester as exchange student in Albuquerque, NM, USA

## LEADERSHIP & SERVICE

- **Hephaestus Applied AI –** *Research Project Head*
- **Bocconi Students for ML –** *Research Member*
- **BSI Bocconi –** *Project Leader*
- **AI & Neuroscience Student Association –** *Member*

## LANGUAGES

**Italian**: Native | **English**: Advanced (IELTS C1 band 8, 2024) | **French**: Beginner (Bocconi course for B1) | **Spanish**: Beginner

*I authorize the processing of personal data present in the CV pursuant to GDPR (EU Regulation 2016/679)*